Title: Understanding Distributions in Financial Markets

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Assignment – Understanding Data

Part 1: Book Chapter Submission

Problem - When we look at how stock prices change over time, do those changes follow a predictable, "normal" pattern or do they behave in unexpected ways that we need to understand?

Relevance and Interest - When you invest money in stocks, the price goes up and down. The **return** is simply how much you gained or lost, usually expressed as a percentage. For example:

- You buy a stock at $100
- It goes up to $105
- Your return is +5%

Now, if we collect thousands of these daily returns and plot them on a chart, what shape do we get?

Many people assume we get a bell curve (also called a "normal distribution") a smooth, symmetric shape where:

- Most returns are small (clustered in the middle)
- Big gains and big losses are rare (at the edges)
- Gains and losses are equally likely

This assumption is baked into many tools that banks, investors, and analysts use every day. But here's the problem: real stock returns don't follow this nice, neat pattern.

In reality:

- **Crashes happen more often than expected.** Events like the 2008 financial crisis or the COVID crash in 2020 are supposed to be extremely rare under the bell curve model — yet they keep happening.
- **Big losses are more common than big gains.** Markets tend to fall fast and rise slowly, creating an uneven pattern.
- **Wild days cluster together.** When markets get volatile, they stay volatile for a while. Calm periods also stick together.

Why should you care?

If you're using a tool that assumes the bell curve, you might think a major crash is a "once in 10,000 years" event. But if crashes happen every 10-20 years, you're dangerously underestimating your risk.

This chapter we will understand:

- How to measure and visualize the "shape" of returns
- How to test whether returns follow a bell curve
- What it means when they don't

1.2 Theory and Background

What is a return?

- A **return** measures how much an investment gained or lost over a period, shown as a percentage.

Simple return formula – Return= Price Today − Price Yesterday/ Price Yesterday ×100

Example: Stock goes from \$100 to \$103 → Return = 3%

What is a Distribution?

A distribution describes how data is spread out. If you calculate daily returns for a stock over 10 years (2,500 data points) and plot them on a histogram, the shape you see is the distribution.

Key questions:

- Where is the center? (mean)
- How spread out is it? (standard deviation)
- Is it symmetric or lopsided? (skewness)
- How often do extreme values occur? (kurtosis)

The Normal Distribution (Bell Curve)

The normal distribution is the most famous distribution in statistics:

- Symmetric around the mean
- Most values cluster near the center
- Extreme values are very rare
- Fully described by just two numbers: mean and standard deviation

Many financial models assume returns follow this pattern. The problem? They often don't.

Four Key Measures -

Measure	
Mean	Average return
SD	How spread-out returns are (volatility)
Skewness	How spread-out returns are (volatility)
Kurtosis	How "fat" are the tails? Higher = more extreme events

A perfect normal distribution has skewness = 0 and kurtosis = 3.

Real stock returns typically show **negative skewness** and **kurtosis > 3** (called "fat tails").

1.3 The Problem

Most risk and investment tools assume stock returns follow a normal (bell curve) distribution. But real returns behave differently:

1. **Fat tails** — Extreme gains and losses happen far more often than the bell curve predicts
2. **Negative skewness** — Large drops are more frequent than large jumps
3. **Volatility clustering** — Turbulent days tend to follow turbulent days

If we ignore these patterns, we underestimate risk.

Input-Output Format

Input:

- Historical stock prices (daily closing prices)
- Example: S&P 500 prices from 2014-2024

Output:

- Daily returns (percentage changes)
- Descriptive statistics: mean, standard deviation, skewness, kurtosis
- Visualizations: histogram, Q-Q plot
- Normality test results (pass/fail)

Sample Data		
Date	**Price**	**Daily Return**
2024-01-01	$100.00	—
2024-01-02	$101.50	+1.50%
2024-01-03	$99.75	-1.72%
2024-01-04	$100.25	+0.50%

1.4 Problem Analysis –

Constraints and Assumptions

Assumptions:

- Stock prices are available daily (no missing data)
- Returns are calculated using closing prices
- We use log returns for mathematical convenience

Constraints:

- Historical data doesn't guarantee future behavior
- Very short time periods may not reveal true distribution shape
- Different assets (stocks, crypto, bonds) may behave differently

Approach to Solving the Problem

Step 1: Collect Data

- Download historical prices using Python (`yfinance` library)

Step 2: Calculate Returns

- Convert prices to daily log returns

Step 3: Compute Statistics

- Mean, standard deviation, skewness, kurtosis

Step 4: Visualize

- Histogram with normal curve overlay
- Q-Q plot (compares data to theoretical normal)

Step 5: Test for Normality

- Jarque-Bera test — checks if skewness and kurtosis match normal distribution
- Shapiro-Wilk test — checks overall fit to normal

Step 6: Interpret Results

- If tests reject normality - returns have fat tails or skewness
- Discuss real-world implications

1.5 Solution Explanation

Step 1: Get the Data

```
import yfinance as yf

data = yf.download("^GSPC", start="2014-01-01", end="2024-01-01") prices = data['Close']
```

Step 2: Calculate Log Returns

```
import numpy as np

returns = np.log(prices / prices.shift(1)).dropna()
```

Step 3: Compute the Four Moments

```
From scipy import stats

mean = np.mean(returns)

std = np.std(returns)
```

```
skewness = stats.skew(returns)

Kurtosis = stats.kurtosis(returns)
```

Step 4: Visualize the Distribution

Create two plots:

- Histogram with normal curve overlay shows actual shape vs. expected
- Q-Q plot if points follow the line, data is normal; if tails curve away, data has fat tails

Step 5: Test for Normality

```
# Jarque-Bera Test (checks skewness and kurtosis)

jb_stat, jb_pvalue = stats.jarque_bera(returns)

# Shapiro-Wilk Test (checks overall fit)

sw_stat, sw_pvalue = stats.shapiro(returns)
```

Interpretation:

- p-value < 0.05 → Reject normality (data is NOT normal)
- p-value ≥ 0.05 → Cannot reject normality

1. LOAD price data from Yahoo Finance

2. CALCULATE daily log returns

3. COMPUTE mean, std, skewness, kurtosis

4. PLOT histogram and Q-Q plot

5. RUN Jarque-Bera and Shapiro-Wilk tests

6. IF p-value < 0.05 THEN reject normality

7. REPORT findings and interpret results

Results and Discussion

1.6 Key Findings

1. All assets reject normality
The Jarque-Bera test returned p-values near zero for all assets, meaning we can confidently say none of them follow a normal distribution.

2. Negative skewness is universal
All assets showed negative skewness, meaning large losses are more common than large gains. Markets crash fast but rise slowly.

3. Fat tails are everywhere
Excess kurtosis was far above zero for all assets:
- Normal distribution: kurtosis = 0
- S&P 500: kurtosis = 16
- Bitcoin: kurtosis = 11

This means extreme events (crashes and spikes) happen much more often than the bell curve predicts.

4. Tail analysis confirms the danger
Events beyond 3 standard deviations should happen about 0.3% of the time under normality. In real data, they happen 3-5x more often.

Discussion

These findings connect directly to the theoretical background. Mandelbrot warned about fat tails in the 1960s, and our analysis confirms his observations still hold today.

Practical implications:
- Risk models using normal distributions will underestimate extreme losses
- Portfolio managers should use fat-tailed distributions for stress testing
- Individual stocks (Tesla) show even more extreme behavior than indices (S&P 500)
- Even "safe" assets like Gold are not normally distributed

Limitations:
- Historical patterns may not predict future behavior
- We only analyzed daily returns (weekly or monthly may differ)
- Results depend on the time chosen

Conclusion:
The assumption of normality in financial returns is convenient but wrong. Investors and risk managers who ignore fat tails do so at their peril.

1.7 References

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